**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**

**Department of Computer Science and Engineering**

**CO2 ASSESSMENT TOOL 1 – SCENARIO-BASED REQUIREMENT ANALYSIS**

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| **Course Code:** CSA10 / CSA1005 | **Course Title:** Software Engineering |
| **CO Assessed:** CO2 | **Assessment:** Assessment Tool 2 – Scenario-Based Requirement Analysis |
| **Weightage:** 20% | **Total Marks:** 25 |
| **CO2 Statement:** Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills. | |
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| **Date:** 27.10.26 | **Duration:** 60 minutes |

## Code of Conduct

I, M RENE DANIEL (Reg. No. 1971260041), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

## Smart Emergency Medicine Availability and Hospital Coordination Platform

**Project Overview**

The Smart Emergency Medicine Availability and Hospital Coordination Platform is a proposed digital system that helps authorized healthcare personnel identify hospitals with suitable emergency facilities, specialist services and critical resources. It brings participating-hospital information into a coordinated platform, supports emergency requests and responses, and maintains a record of coordination activities. The platform is intended to reduce delays caused by repeated manual calls and fragmented information while protecting sensitive information through access controls.

## 1. Industry Context

During medical emergencies, patients and healthcare providers may need to locate a hospital that can provide the required treatment, specialist service, equipment, medicine or critical-care facility. Information may be spread across telephone calls, spreadsheets and separate hospital systems. Since beds, specialists, medicines and equipment can become unavailable quickly, staff may struggle to obtain a current and reliable picture of service availability. A coordinated platform can improve visibility by maintaining hospital profiles, facility details and resource updates, with timestamps showing when information was last confirmed.

## 2. Problem Statement

Design and develop a Smart Emergency Hospital Coordination Platform that maintains real-time or regularly updated information about participating hospitals, available departments, emergency services and critical resources. The system should allow authorized healthcare personnel to identify potentially suitable facilities based on patient requirements and reported availability. It should provide hospital dashboards, emergency request management, facility information, secure communication, coordination histories and operational reports while ensuring appropriate privacy and access controls. Since availability may change, the platform must clearly indicate whether information is current, stale or awaiting confirmation.

## 3. Aim

To improve emergency healthcare coordination by providing a secure and centralized way for authorized personnel to search hospital capabilities, submit emergency requests, obtain hospital responses and track coordination progress.

## 4. Objectives

- Maintain participating hospital profiles, locations and authorized contact information.
- Record emergency facilities and services, including emergency departments, intensive care, trauma care, imaging, laboratories and operating theatres where applicable.
- Maintain department, specialist category, service-hour and on-call information where provided by hospitals.
- Track selected critical resources such as bed categories, oxygen support, blood products, equipment and emergency medicines where reliable updates are supported.
- Receive emergency coordination requests with urgency, required services and minimum necessary patient information.
- Match requests to potentially suitable hospitals using required facilities, reported availability and configured location criteria.

- Notify authorized hospital personnel of new requests, responses and status changes.
- Maintain time-stamped coordination histories and secure communication records.
- Generate operational reports on requests, response times, resource updates and coordination outcomes.
- Protect sensitive data through authentication, role-based access, audit logs and privacy controls.

## 5. Scope of the System

### Included

- Hospital registration and profile management.
- Facility, department, specialist and service directory.
- Resource status updates with last-updated and verification indicators.
- Emergency request submission, matching, assignment and status tracking.
- Notifications and secure communication between authorized personnel.
- Role-specific dashboards, coordination history and operational reporting.

### Not Included in the Initial Version

- Automated diagnosis or replacement of clinical judgment.
- Guaranteeing that a displayed resource will remain available until a patient arrives.
- Integration with every hospital system without approval or technical support.
- Replacement of ambulance dispatch, official emergency channels or local clinical protocols.

## 6. Stakeholders and Actors

- **Emergency Coordinator / Referring Healthcare Professional:** creates requests, specifies required services, reviews candidate hospitals and tracks progress.
- **Hospital Administrator:** manages hospital profiles, departments, facilities, users and permissions.
- **Hospital Availability Officer:** updates resource information and confirms reported availability.
- **Receiving Hospital Clinician or Authorized Staff:** reviews requests and accepts, declines or requests clarification.
- **System Administrator:** manages platform configuration, accounts, access policies and technical operation.
- **Reporting or Audit Officer:** reviews permitted operational reports and audit histories.
- **Notification Service:** delivers in-platform alerts and, where configured, email or SMS notifications.

## 7. Functional Requirements

- **FR-01 Hospital Management:** authorized administrators can create, update, verify and deactivate hospital profiles.
- **FR-02 Facility Directory:** the system stores facilities and emergency services offered by each hospital.
- **FR-03 Department and Specialist Management:** authorized staff maintain department, specialty, service-hour and on-call details.
- **FR-04 Resource Updates:** permitted users update supported resource status or quantity, including the update time and responsible user.
- **FR-05 Emergency Request Creation:** users submit a request with unique ID, urgency, required services, requesting facility and necessary contact details.
- **FR-06 Hospital Matching:** the system filters hospitals using required services and available information, and displays data freshness.
- **FR-07 Hospital Responses:** receiving personnel can accept, decline or request clarification, with a response note.
- **FR-08 Status Tracking:** requests move through defined states such as Submitted, Under Review, Response Received, Accepted, Declined, Escalated, Closed or Cancelled.
- **FR-09 Notifications:** designated users receive alerts for new requests, responses and important status changes.
- **FR-10 Secure Communication:** the system records request-related messages for permitted users.
- **FR-11 Search and Filters:** users can search by hospital, location, service, department, specialist or supported resource.
- **FR-12 History and Audit:** important actions are recorded with user, timestamp and action details.

- **FR-13 Dashboards:** role-specific dashboards show open requests, pending responses and relevant resource information.
- **FR-14 Reports:** authorized users can generate reports on request volumes, response intervals and data freshness.
- **FR-15 Access Control:** authentication and role-based permissions restrict access to authorized users and records.
- **FR-16 Stale-Data Warnings:** information older than a configurable period is marked as stale and requires confirmation.
- **FR-17 Request Closure:** authorized users close or cancel requests with a reason while preserving the history.

## 8. Non-Functional Requirements

- **Performance:** searches and dashboards should respond promptly under expected operating load.
- **Availability and Reliability:** provide monitoring, backups, recovery procedures and handling for failed notifications.
- **Security:** use secure authentication, role-based authorization, encrypted network communication and security logging.
- **Privacy:** collect only information needed for coordination and restrict access by role and purpose.
- **Usability:** present clear status labels and simple forms suitable for time-sensitive workflows.
- **Scalability:** support additional hospitals, users, services and resource categories.
- **Interoperability:** support documented interfaces for approved hospital-system integrations.
- **Data Accuracy:** display source, update time and confirmation state; do not present unverified availability as guaranteed.
- **Maintainability:** use modular components, documentation and testing.
- **Auditability:** ensure important changes can be traced to an authenticated user.

## 9. Main Use Cases and Workflow

1. Authorized user signs in and creates an emergency coordination request.
2. User enters urgency, required facilities or specialist services and only the patient information necessary for coordination.
3. The platform validates the request and searches participating hospitals against the required services.
4. Candidate hospitals are shown with reported resource status, location information where configured, and last-updated timestamps.
5. The coordinator sends the request to appropriate hospital contacts.
6. Receiving personnel review the request and accept, decline or request clarification.
7. The system records responses, updates status and notifies relevant authorized users.
8. The coordinator follows local protocols to confirm arrangements directly with the receiving hospital.
9. The request is closed with an outcome, and its permitted history can support audit and reporting.

## 10. Data Requirements

- **Hospital:** Hospital ID, name, address or region, contact details, participation status and verification date.
- **User:** User ID, role, affiliated hospital, account status and access permissions.
- **Department:** Department ID, hospital ID, name, service hours and status.
- **Specialist Service:** specialty category, service window and reported availability.
- **Facility:** facility type, supported service, operating status and notes.
- **Resource Status:** resource type, reported quantity or state, updated-by user, timestamp and verification state.
- **Emergency Request:** request ID, requester, urgency, required services, minimum necessary patient details, creation time and current status.
- **Hospital Response:** request ID, hospital ID, response status, note and timestamp.
- **Notification and Audit Event:** recipient, channel, delivery status, actor, action, target record and timestamp.

## 11. Privacy, Assumptions and Constraints

- Only authenticated, authorized users should access coordination records and sensitive patient information.
- Participating hospitals are expected to maintain their service and resource information and assign staff to respond.
- Real-time availability is possible only when supported by reliable updates or approved technical integrations.
- Resource availability can change rapidly; all results must show freshness and require direct confirmation for final arrangements.

- Network failures or unavailable integrations may interrupt coordination; an alternative contact procedure should be available.
- Applicable laws, institutional policies and data-retention requirements must guide storage and sharing of patient information.
- The platform supports coordination and does not replace clinical judgment, emergency dispatch or established protocols.

## 12. Requirement Validation and Acceptance Criteria

Requirements should be reviewed with emergency coordinators, hospital representatives, clinicians, administrators and IT/security personnel. Each requirement should be checked for clarity, feasibility, consistency, testability and traceability. The system should be accepted when:

- Authorized administrators can maintain hospital and facility profiles.
- Authorized staff can update supported resources and see update timestamps.
- Coordinators can create a request and receive a unique request ID.
- Search results reflect required services and clearly identify stale or missing data.
- Receiving hospitals can respond, and permitted users can view response status.
- Notifications and request histories are recorded where supported.
- Users cannot access records outside their permissions.
- Authorized users can produce basic operational reports and audit records.

## 13. Expected Outcomes

- Faster identification of potentially suitable healthcare facilities.
- Improved emergency coordination between referring and receiving hospitals.
- Reduced communication delays caused by repeated manual calls.
- Better visibility of available services, departments and reported critical resources.
- More structured emergency request handling and response tracking.
- Improved traceability through time-stamped coordination histories.
- More informed operational planning through reports on request patterns and response times.
- More efficient resource utilization where participating hospitals provide accurate and timely updates.

## 14. Conclusion

The Smart Emergency Medicine Availability and Hospital Coordination Platform provides a structured approach to sharing hospital capability information and coordinating emergency requests. Its effectiveness depends on participation by hospitals, reliable updates, secure access, clear workflows and direct confirmation of availability. A phased implementation can begin with hospital profiles, service directories and request tracking, then expand to resource integrations and analytics. Clinical decisions remain the responsibility of qualified healthcare professionals.

## Rubrics for Calculation

The following assessment rubric is retained from the original assessment document.

| Criteria | Excellent (4 Marks) | Good (3 Marks) | Satisfactory (2 Marks) | Needs Improvement (1 Mark) |
|---|---|---|---|---|
| **Scenario Understanding and Problem Analysis (30%)** | Thorough understanding of the scenario, stakeholders, objectives and business context with clear analysis. | Good understanding with minor omissions. | Basic understanding with limited analysis. | Poor understanding; major aspects missing or incorrect. |
| **Functional and Non-Functional Requirement Identification (25%)** | Comprehensive, accurate and well-classified functional and non-functional requirements. | Most requirements identified with minor classification errors. | Only basic requirements; several omissions. | Requirements incomplete, inaccurate or poorly classified. |
| **Actor Identification and Use Case Modelling (20%)** | All relevant actors and use cases identified; complete, accurate UML use case diagram. | Most actors and use cases identified; diagram has minor errors. | Limited actors/use cases; diagram partially correct. | Actors/use cases missing or incorrect; diagram absent or inaccurate. |
| **Software Requirement Specification (SRS) Documentation (15%)** | Well-structured, complete and professionally organized SRS with all required sections. | Organized SRS with minor missing sections or formatting issues. | Essential sections only, with limited detail. | Incomplete or poorly organized; essential sections missing. |
| **Assumptions, Constraints and Requirement Validation (10%)** | Realistic assumptions and constraints; validates completeness, consistency and ambiguity with justification. | Most assumptions and constraints identified; validation adequate. | Limited assumptions/constraints; superficial validation. | Assumptions, constraints and validation missing or incorrect. |

## Marks

| Criteria | Mark |
|---|---|
| Scenario Understanding and Problem Analysis (30%) | /5 |
| Functional and Non-Functional Requirement Identification (25%) | /5 |
| Actor Identification and Use Case Modelling (20%) | /5 |
| Software Requirement Specification (SRS) Documentation (15%) | /5 |
| Assumptions, Constraints, and Requirement Validation (10%) | /5 |
| **TOTAL** | **/25** |